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United States Patent	12413066
Kind Code	B2
Date of Patent	September 09, 2025
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Control device, pump unit, and refrigerant circulation device

Abstract

A technology of starting power supply from an external device to a load device in a shorter period, after detection of connection between the external device and the load device, is provided. A control device includes a first connector attachable to and detachable from an external device, a power supply path to electrically connect a load device and the first connector, a load switch to switch between connection and disconnection of the power supply path, a protection circuit to protect the load switch from an inrush current to the load switch, the protection circuit including at least a capacitor, and a control circuit. The control circuit is operable to switch the load switch from disconnection to connection after a total time of a preset set time and a first time based on a charging time of the capacitor which has elapsed since connection of the first connector to the external device was detected.

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Appl. No.: **18/372269**

Filed: **September 25, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240114655 A1	Apr. 04, 2024

Foreign Application Priority Data

JP	2022-156324	Sep. 29, 2022

Publication Classification

Int. Cl.: **H02M7/44** (20060101); **H02H3/093** (20060101); **H02H7/085** (20060101); **H02H9/00** (20060101)

U.S. Cl.:

CPC **H02H3/093** (20130101); **H02H7/085** (20130101); **H02H9/005** (20130101);

Field of Classification Search

CPC: H02H (3/093); H02H (7/085); H02H (9/005); H02H (9/004); F05D (2270/52); H05K (7/20272)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2022-156324, filed on Sep. 29, 2022, the entire contents of which are hereby incorporated herein by reference.

1. Field of the Invention

(2) The present disclosure relates to a control device, a pump unit, and a refrigerant circulation device.

2. Background

(3) Conventionally, a hot plug function of a disk drive in a redundant array of inexpensive disks (RAID) system is known. In a RAID system, a power-on delay circuit controls power supply to a disk drive when the disk drive is inserted into an array. Specifically, the power-on delay circuit detects that connection is established between the hard drive and the backplane connector of the array. When a connection is first sensed, the power-on delay circuit starts clocking a predetermined time (that is, timeout). Power to the disk drive is applied via a solid state switch only when a predetermined time of the power-on delay circuit has elapsed.
(4) Conventionally, power is supplied from an external device (that is, RAID system) to a load device (that is, disk drive) in response to elapse of a predetermined time from initial detection of connection between the external device and the load device. The length of the predetermined time is required to be as short as possible, but the length of the predetermined time is not known.
SUMMARY
(5) A control device according to one example embodiment of the present disclosure includes a first connector, a power supply path, a load switch, a protection circuit to protect the load switch from an inrush current to the load switch, and a control circuit. An external device is attachable to and detachable from the first connector. The power supply path electrically connects a load device and the first connector. The load switch is switchable between connection and disconnection of the power supply path. The protection circuit includes at least a capacitor. The control circuit is operable to switch the load switch from disconnection to connection after a total time of a preset set time and a first time based on a charging time of the capacitor which has elapsed since connection of the first connector to the external device was detected.
(6) A pump assembly according to another example embodiment of the present disclosure includes a first connector, a motor, a pump rotor, a power supply path, a load switch, a protection circuit to protect the load switch from an inrush current to the load switch, and a control circuit. The first connector is attachable to and detachable from an external device. The pump rotor is rotatable by power generated by the motor. The power supply path is electrically connected between the motor and the first connector. The load switch is operable to switch between connection and disconnection of the power supply path. The protection circuit includes at least a capacitor. The control circuit is operable to switch the load switch from disconnection to connection after the total time of a preset set time and a first time based on the charging time of the capacitor has elapsed since connection of the first connector to the external device was detected. The motor generates the power in response to the load switch being switched to connection.
(7) A refrigerant circulation device according to still another example embodiment of the present disclosure includes a pump assembly and a flow path. In the flow path, the refrigerant flows by rotation of the pump rotor.
(8) The above and other elements, features, steps, characteristics and advantages of the present disclosure will become more apparent from the following detailed description of the example embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a block diagram illustrating a configuration of a cooling system according to an example embodiment of the present disclosure.
(2) FIG. **2** is a schematic view illustrating insertion and removal of a casing and each pump assembly according to an example embodiment of the present disclosure in detail.
(3) FIG. **3** is a block diagram illustrating a detailed configuration of a pump assembly illustrated in FIGS. **1** and **2**.
(4) FIG. **4** is a timing chart illustrating an operation when the pump assembly is attached to the

casing in the refrigerant circulation device illustrated in FIG. **1**.
(5) FIG. **5** is a timing chart illustrating an operation when the main power switch of the casing is turned off in the refrigerant circulation device illustrated in FIG. **1**.
(6) FIG. **6** is a timing chart illustrating an operation when the pump assembly is removed from the casing in the refrigerant circulation device illustrated in FIG. **1**.
(7) FIG. **7** is a timing chart illustrating operation of the pump assembly according to an example embodiment of the present disclosure when an operator attempts to remove the pump assembly from the casing and then reattaches the pump assembly to the casing.

DETAILED DESCRIPTION

(8) Hereinafter, example embodiments of the present disclosure will be described with reference to the drawings. In the drawings, the same or corresponding parts are denoted by the same reference numerals, and the description will not be repeated.
(9) FIG. **1** is a block diagram illustrating a configuration of a cooling system **100** according to an example embodiment of the present disclosure. The cooling system **100** includes a cooling device **1** and a refrigerant circulation device **2**.
(10) The cooling device **1** includes a distribution manifold **11**, a plurality of cold plates **12**, a plurality of heat sources **13**, and a collection manifold **14**. The number of each of the cold plates **12** and the heat sources **13** may be at least one.
(11) In the cooling system **100**, the refrigerant circulates among the refrigerant circulation device **2**, the distribution manifold **11**, the plurality of cold plates **12**, and the collection manifold **14** as indicated by a plurality of arrows A**01** to A**05**. The refrigerant is, for example, a coolant. Examples of the coolant include antifreeze liquid and pure water. A typical example of antifreeze liquid is an ethylene glycol aqueous solution or a propylene glycol aqueous solution. High-temperature refrigerant flows into the refrigerant circulation device **2** from the collection manifold **14** (see arrow A**01**). The refrigerant circulation device **2** pressurizes and cools the refrigerant. When the refrigerant is pressurized, the refrigerant circulates in the cooling system **100** (see arrows A**01** to A**05**). Specifically, the low-temperature refrigerant flows into the plurality of cold plates **12** via the distribution manifold **11** (see arrow A**04**) and flows through the plurality of cold plates **12**. The plurality of cold plates **12** are in thermal contact with the plurality of heat sources **13**. Each heat source **13** is a device that generates heat. In the example embodiments, each heat source **13** is a component of a computing device. Examples of the heat source **13** include an electrolytic capacitor, a power semiconductor module, and a printed circuit board.
(12) Each of the cold plates **12** has an inflow port **121** and an outflow port **122**. In FIG. **1**, for convenience, reference numerals “**121**” and “**122**” are representatively given to only one cold plate **12**. The refrigerant flows into each of the inflow ports **121** from the downstream end **111** of the distribution manifold **11** (see arrow A**04**). The refrigerant flows from the inflow port **121** toward the outflow port **122** at each cold plate **12**. Accordingly, the heat generated by the heat source **13** moves to the refrigerant flowing in each cold plate **12**. That is, the temperature of the refrigerant becomes high. The high-temperature refrigerant flows out from each outflow port **122** to each upstream end **141** of the collection manifold **14** and flows in the collection manifold **14** (see arrow A**05**).
(13) The refrigerant circulation device **2** includes a casing **21**, two pump assemblies **22**, a cooling unit **23**, a flow path **24** including pipes **241** to **243**, a power supply unit **25**, and a control circuit **26**. A portion of the refrigerant circulation device **2** excluding the pump assembly **22** is an example of an “external device” of the present disclosure.
(14) The casing **21** has an inflow port **211** for refrigerant and an outflow port **212** for refrigerant. The inflow port **211** is connected with a downstream end **142** of the collection manifold **14**. The refrigerant flows into the inflow port **211** from the downstream end **142**. The outflow port **212** is connected with an upstream end **112** of the distribution manifold **11**. The refrigerant flows out from the outflow port **212** to the upstream end **112**.

(15) The casing **21** accommodates the two pump assemblies **22** and the cooling unit **23**. The cooling unit **23** and the respective pump assemblies **22** are connected between the inflow port **211** and the outflow port **212** by the pipes **241** to **243**. As a result, in the casing **21** (that is, the cooling device **1**), the refrigerant can flow from the inflow port **211** to the outflow port **212** via the cooling unit **23** and the respective pump assemblies **22** sequentially (see arrows A**01** to A**03**).
(16) Each pump assembly **22** has a suction port **221**, a discharge port **222**, and a pump rotor **223** to pressurize the refrigerant in the pipes **241** to **243**. The suction port **221** is connected to the downstream end of the pipe **242**. The discharge port **222** is connected to the upstream end of the pipe **243**. In the pump assembly **22**, the pump rotor **223** rotates to apply pressure to the refrigerant in the pump assembly **22**. As a result, the refrigerant in the pipe **242** is sucked from the suction port **221**. The sucked refrigerant is discharged from the discharge port **222** to the pipe **243**.
(17) The type of the pump assembly **22** is not particularly limited. That is, as the pump assembly **22**, for example, a centrifugal pump, a propeller pump, a viscous pump, or a rotary pump can be adopted. The pump rotor **223** is an impeller when the pump assembly **22** is a centrifugal pump, a propeller pump, a viscous pump, or a gear pump. The pump rotor **223** is a screw when the pump assembly **22** is a screw pump. The number of pump assemblies **22** may be at least one.
(18) The cooling unit **23** cools the refrigerant flowing in the refrigerant circulation device **2**. The type of the cooling unit **23** is not particularly limited. That is, as the cooling unit **23**, an air cooling system or a water cooling system can be adopted. In the case of the air cooling system, the cooling unit **23** includes a radiator and a fan. The radiator is connected to the downstream end of the pipe **241**. High-temperature refrigerant flows into the radiator from the downstream end of the pipe **241**. The radiator is connected to the upstream end of the pipe **242**. The radiator guides the refrigerant flowing in from its own inflow port to its own outflow port. In the process, the refrigerant flowing in the radiator is cooled by the airflow generated by the fan. As a result, low-temperature refrigerant flows out from the outflow port of the radiator.
(19) The power supply unit **25** is a power supply circuit or the like, and generates, for example, a DC voltage Vcc from an AC voltage supplied from, for example, a commercial power supply. The value of the DC voltage Vcc is not particularly limited, and is, for example, 54 V. The power supply unit **25** supplies the generated DC voltage Vcc to each pump assembly **22** and the cooling unit **23**.
(20) The control circuit **26** includes a microcomputer, a memory, and the like, not illustrated. The microcomputer operates according to a program stored in the memory and controls the operation of each pump assembly **22** and the cooling unit **23**.
(21) FIG. **2** is a schematic view illustrating insertion and removal of a casing **21** and each pump assembly **22** in detail. As illustrated in FIGS. **1** and **2**, the casing **21** has a predetermined shape. The predetermined shape is, for example, a substantially rectangular parallelepiped shape. On a first surface **213** of the casing **21**, the number of openings **214** corresponding to the number of the pump assemblies **22** is formed. An accommodation space **215** is formed from each opening **214** toward the inside of the casing **21**. Each pump assembly **22** is movable by a human hand in the approaching direction D**01** and the separation direction D**02** in the accommodation space **215** through the opening **214**. The approaching direction D**01** is a direction from the opening **214** toward a mounting position P**01** defined in advance in the accommodation space **215**. The separation direction D**02** is a direction opposite to the approaching direction D**01** and is a direction from the mounting position P**01** toward the opening **214**. The downstream end of the pipe **242**, the upstream end of the pipe **243**, and the connector **216** are disposed at the back of the accommodation space **215**. The connector **216** is an example of a “second connector” in the present disclosure. Note that details of the connector **216** will be described later.
(22) The pump assembly **22** has a shape corresponding to the opening **214** and the accommodation space **215**, that is, a substantially rectangular parallelepiped shape. In a state where each pump assembly **22** is located at the mounting position P**01**, the suction port **221** of the pump assembly **22**

is connected to the downstream end of the pipe **242**, and the discharge port **222** of the pump assembly **22** is connected to the upstream end of the pipe **243**. Each pump assembly **22** is further fixed to the first surface **213** with a fixing tool such as a screw (not illustrated) in a state of being located at the mounting position P**01**. As a result, the refrigerant can flow from the pipe **242** to the suction port **221**, and the refrigerant can flow from the discharge port **222** to the pipe **243**. Since each pump assembly **22** is fixed to the first surface **213**, the pipe **242** is prevented from coming out from the suction port **221** and the pipe **243** is prevented from coming out from the discharge port **222**.

(23) Each pump assembly **22** further includes a connector **224** attachable to and detachable from the connector **216** (that is, an external device). The connector **224** is an example of a "first connector" in the present disclosure. When the pump assembly **22** moves in the approaching direction D**01** and is located at the mounting position P**01**, the connector **224** is electrically connected to the connector **216**. While the pump assembly **22** moves from the mounting position P**01** in the separation direction D**02**, the connector **224** is removed from the connector **216**. Note that the details of the connector **224** will be described later.

(24) FIG. **3** is a block diagram illustrating a detailed configuration of the pump assembly **22** illustrated in FIGS. **1** and **2**.

(25) FIG. **3** illustrates the connector **216** and the power supply unit **25** in addition to the detailed configuration of the pump assembly **22**. The connector **216** includes at least terminals **216**A to **216**C. The DC voltage Vcc generated by the power supply unit **25** is applied between the terminals **216**A and **216**B. Among the terminals **216**A to **216**C, the terminals **216**B and **216**C are grounded. Specifically, the terminals **216**B and **216**C are electrically connected to the ground in the power supply unit **25**. Note that the terminal **216**C is an example of a "second detection terminal" in the present disclosure.

(26) As illustrated in FIG. **3**, the pump assembly **22** includes a power supply unit **2213**, a drive unit **225**, a motor **226**, a control circuit **227**, a power supply path **228**, a load switch **229**, a protection circuit **2210**, a drive unit **2211**, and a control circuit **2212**, in addition to the pump rotor **223** and the connector **224** described above. At least the connector **224**, the power supply path **228**, the load switch **229**, the protection circuit **2210**, and the control circuit **2212** constitute a control device **27**.

(27) The connector **224** includes at least terminals **224**A to **224**C. The terminals **224**A, **224**B, and **224**C are electrically connected to the terminals **216**A, **216**B, and **216**C when the connectors **216** and **224** are electrically connected. Here, the timing at which the terminals **216**B and **224**B conduct in the process of connecting the connectors **216** and **224** to each other substantially coincides with the timing at which the terminals **216**A and **224**A conduct. On the other hand, for example, since the terminal **216**C is formed in a shape different from that of the terminal **216**A, the timing at which the terminals **216**C and **224**C conduct is delayed by a predetermined time from the timing at which the terminals **216**A and **224**A conduct. Note that the terminal **224**C is an example of a "first detection terminal" in the present disclosure.

(28) The power supply unit **2213** is a power supply circuit or the like, and generates a DC voltage Vdd from the DC voltage Vcc supplied from the terminal **224**A. The value of the DC voltage Vdd is not particularly limited, but is lower than the withstand voltage of each microcomputer of the control circuits **227** and **2212**. The DC voltage Vdd is lower than the DC voltage Vcc, for example, 3.3 V. The DC voltage Vdd is supplied to the control circuit **2212**. The control circuit **2212** operates with the DC voltage Vdd. Note that the power supply unit **2213** may be a battery that outputs the DC voltage Vdd instead of the power supply circuit.

(29) The drive unit **225** is, for example, an H-bridge circuit. The drive unit **225** includes terminals **225**A and **225**B. In the drive unit **225**, a drive voltage based on the DC voltage Vcc is applied between the terminals **225**A and **225**B through the connector **224** or the like, in response to the elapse of a predetermined time after the connector **216** and **224** is electrically connected. In the H-bridge circuit, the four switching elements are turned on and off under the control by the control

circuit **227**. As a result, the drive unit **225** controls the direction of the current flowing through the motor **226** and the rotation speed of the motor **226**.

(30) The motor **226** has a rotatable output shaft. The pump rotor **223** is mechanically connected to the output shaft. The motor **226** rotates under the control of the drive unit **225** to generate power. The motor **226** is an example of a "load device" of the present disclosure. As is well known, the motor **226** detects the rotation speed of the output shaft, and outputs a signal indicating the detected rotation speed (hereinafter, it is simply referred to as "rotation speed") to the control circuit **227**.

(31) The pump rotor **223** rotates by the power generated by the motor **226**.

(32) The control circuit **227** includes a microcomputer, a memory, and the like, not illustrated. The microcomputer operates according to a program stored in the memory. Specifically, the control circuit **227** outputs the rotational speed input from the motor **226** to the control circuit **2212**. In addition, the control circuit **227** turns on and off each switching element included in the H-bridge circuit on the basis of a pulse width modulation (PWM) signal output from the control circuit **2212**. The PWM signal is an example of a "pulse signal" in the present disclosure. Furthermore, the control circuit **227** may be integrated with the control circuit **2212**.

(33) The power supply path **228** electrically connects the motor **226** (that is, a load device) and the connector **224**. Specifically, the power supply path **228** includes two power lines **228**A and **228**B. The power line **228**A electrically connects the terminals **224**A and **225**A. The power line **228**B electrically connects the terminals **224**B and **225**B.

(34) The load switch **229** switches between connection and disconnection of the power supply path **228**. Specifically, the load switch **229** is provided on the power line **228**A. The load switch **229** typically includes a metal oxide semiconductor field effect transistor (MOSFET). In the MOSFET, the source is disposed on the input side of the DC voltage Vcc. That is, the source is electrically connected to the terminal **224**A. The drain is disposed on the output side of the DC voltage Vcc. That is, the drain is connected to the Vcc terminal of the drive unit **225**. The gate is electrically connected to a drive unit **2211** described later. In response to a high-level switch signal being output from the control circuit **2212** to the drive unit **2211**, a current flows between the collector and the emitter of the NPN transistor in the drive unit **2211**, and as a result, a current flows between the source and the drain of the load switch **229**.

(35) The protection circuit **2210** protects the load switch **229** from an inrush current that may flow through the load switch **229** when the connectors **224** and **216** are connected to each other. The protection circuit **2210** may include at least a capacitor. In the example embodiment, the protection circuit **2210** includes a diode, a capacitor, and a resistor. The diode, the capacitor, and the resistor are all connected between the connector **224** and the load switch **229** and between the load switch **229** and the drive unit **2211** in the power line **228**A.

(36) The drive unit **2211** controls on/off of the load switch **229** (that is, MOSFET). Specifically, the drive unit **2211** includes an NPN transistor. In the NPN transistor, the collector is electrically connected to the gate of the MOSFET via the resistor. The emitter is electrically connected to the power line **228**B. The base is electrically connected to a terminal **2212**B in the control circuit **2212**.

(37) Note that the drive unit **2211** can also be realized using a PNP transistor and a photocoupler, as is well known, besides the NPN transistor.

(38) The control circuit **2212** includes a microcomputer, a memory, and the like, not illustrated. The microcomputer has at least terminals **2212**A to **2212**E. The microcomputer operates by the DC voltage Vdd supplied to the terminal **2212**E. The operation of the microcomputer is defined by a program stored in the memory.

(39) The terminal **2212**A is an input terminal of a hot plug signal (hereinafter referred to as an "HP signal"). The terminal **2212**A is electrically connected to the terminal **224**C. In addition, a DC voltage Vdd generated by the power supply unit **2213** is supplied to the terminal **2212**A via a pull-up resistor **2212**F. Therefore, when the terminals **216**C and **224**C are not electrically connected to each other (that is, in a case where the pump assembly **22** is not mounted on the casing **21**), the DC

voltage Vdd is input to the terminal **2212**A. That is, an HP signal is at a high level (DC voltage Vdd). On the other hand, when the terminals **216**C and **224**C are electrically connected to each other, the terminal **2212**A is connected to the ground of the power supply unit **25**. That is, an HP signal is at a low level (0 V).
(40) The terminal **2212**B is an output terminal for a switch signal (hereinafter referred to as an "SW signal"). In the control circuit **2212**, the microcomputer incorporates a timer. The timer may be an integrated circuit externally attached to the microcomputer. The timer starts clocking with transition of the HP signal from a high level to a low level as a trigger. The microcomputer outputs a high-level SW signal from the terminal **2212**B in response to the value of the timer reaching a set time having been set.
(41) The terminal **2212**C is an input terminal for the rotation speed. The terminal **2212**D is an output terminal for a PWM signal. The microcomputer outputs, to the control circuit **227**, a pulse width modulation (PWM) signal whose pulse width is adjusted so that the actual rotation speed of the motor **226** approaches the target rotation speed of the motor **226**.
(42) FIG. **4** is a timing chart illustrating an operation when the pump assembly **22** is mounted on the casing **21** in the refrigerant circulation device **2** illustrated in FIG. **1**.
(43) As illustrated in FIG. **4**, it is assumed that the pump assembly **22** is not mounted on the casing **21** before time t**01**. That is, the connector **216** is not connected to the connector **224** (see FIG. **2**). Therefore, the DC voltage Vcc is not applied between the terminals **224**A and **224**B (see FIG. **3**). The DC voltage Vdd is 0 V. No drive signal is input to the motor **226** (that is, a load device). Both the SW signal and the HP signal are at a low level. Therefore, the load switch **229** does not connect the power line **228**A. Further, at time t**01**, the control circuit **2212** does not start clocking by the timer and does not output a PWM signal.
(44) Before time t**01**, it is assumed that a main power switch (not illustrated) provided in the casing **21** is operated by an operator. Therefore, in the connector **216**, the DC voltage Vcc generated by the power supply unit **25** appears between the terminals **216**A and **216**B.
(45) At time t**01**, the operator inserts the pump assembly **22** into the accommodation space **215** of the casing **21** and moves the pump assembly **22** to the mounting position P**01** (see FIG. **2**). Therefore, among the terminals **224**A, **224**B, and **224**C (see FIG. **3**), the terminals **224**A and **224**B are electrically connected to the terminals **216**A and **216**B, respectively. As a result, the DC voltage Vcc is applied between the terminals **224**A and **224**B. In addition, the power supply unit **2213** generates a DC voltage Vdd from the DC voltage Vcc and provides the DC voltage Vdd to the control circuits **227** and **2212**. The control circuits **227** and **2212** start operation with the DC voltage Vdd generated by the power supply unit **2213**. Thereafter, the control circuit **227** waits for input of a PWM signal from the control circuit **2212** and the rotation speed from the motor **226**. The control circuit **2212** starts monitoring of the HP signal. The HP signal goes to a high level after the DC voltage Vdd is input until the terminals **216**C and **224**C are electrically connected to each other.
(46) After time t**01**, it is assumed that the terminals **216**C and **224**C are electrically connected at time t**02**. The control circuit **2212** detects the conduction of the terminals **216**C and **224**C (that is, the connector **224** is connected to the connector **216**) based on the voltage value of the terminal **2212**A. Specifically, when the terminals **216**C and **224**C are electrically connected, the HP signal transitions from a high level to a low level. The control circuit **2212** recognizes that the HP signal has transitioned from a high level to a low level based on the voltage of the terminal **2212**A. With this recognition, the control circuit **2212** detects that the connector **224** is connected to the connector **216**. In this manner, the control circuit **2212** can easily detect attachment and detachment of the connector **224** to and from the connector **216**. In addition, the control circuit **2212** starts clocking by a timer with detection of connection as a trigger. The control circuit **2212** periodically monitors the elapsed time measured by the timer.
(47) After time t**02**, at time t**03**, when the control circuit **2212** recognizes that the timer has reached

a set time ("20" in the drawing), the control circuit outputs a high-level SW signal from the terminal **2212**B. The drive unit **2211** gives a drive signal to the gate of the load switch **229** in response to the input of the high-level SW signal. However, the protection circuit **2210** is provided in front of the load switch **229**. Therefore, immediately after the drive signal is applied to the load switch **229**, a current flows through the capacitor of the protection circuit **2210**, and the capacitor is charged. That is, the load switch **229** is slowly switched from disconnection to connection. As a result, an inrush current is prevented from flowing through the load switch **229** and the drive unit **225**. As a result, the power supply unit **25** is prevented from stopping or the drive unit **225** is prevented from being damaged by the spark.
(48) A time at which the first time has elapsed from time t**03** is defined as t**04**. The first time is a predetermined time, and is a time based on the charging time of the capacitor of the protection circuit **2210**. Specifically, the first time is the charging time itself of the capacitor, a time slightly shorter than the charging time of the capacitor, or a time slightly longer than the charging time of the capacitor. When the capacitor is fully charged or nearly fully charged at time t**04**, the load switch **229** is switched to connection. As a result, a DC voltage Vcc is applied between the gate and the source of the MOSFET of the load switch **229** and between the terminals **225**A and **225**B of the drive unit **225**. That is, in the load switch **229**, after the control circuit **2212** detects that the connector **224** is connected to the connector **216** (that is, an external device), the load switch **229** is switched from disconnection to connection after the total time of the set time and the first time has elapsed. The first time is a time based on the charging time of the capacitor. As a result, after the connection with the external device is detected, the power supply from the external device to the component (that is, the motor **226**) is started in a shorter time. Specifically, it is possible to relatively shorten the time during which the load switch **229** is disconnected while preventing an inrush current from flowing. As a result, power supply to the motor **226** is started relatively early.
(49) In addition, the load switch **229** is in a disconnected state during the total time. As a result, power can be supplied to the motor **226** after the connection state between the connectors **216** and **224** is stabilized.
(50) A time at which the second time has elapsed from time t**03** is defined as t**05**. The second time is a preset time. Specifically, the second time is shorter than the set time and slightly longer than the first time. When recognizing that the second time has elapsed, the control circuit **2212** starts outputting of a PWM signal to the control circuit **227**. Since the start of outputting of the PWM signal is delayed from the connection of the load switch **229**, the operation of the motor **226** is stabilized. As a result, the motor **226** is driven by the drive unit **225** to start rotation of the pump rotor **223**. Thereafter, the control circuit **2212** outputs, to the control circuit **227**, the PWM signal whose pulse width is adjusted based on the rotation speed from the control circuit **227**.
(51) In addition, since the second time is shorter than the set time, the power supply to the motor **226** is started relatively early.
(52) FIG. **5** is a timing chart illustrating an operation when the main power switch of the casing **21** is turned off in the refrigerant circulation device **2** illustrated in FIG. **1**.
(53) As illustrated in FIG. **5**, it is assumed that the pump assembly **22** is mounted on the casing **21** before time t**11**. That is, the connector **216** is connected to the connector **224** (see FIG. **2**). Therefore, the DC voltage Vcc is applied between the terminals **224**A and **224**B (see FIG. **3**). The DC voltage Vdd is 3.3 V. It is also assumed that a drive signal to the motor **226** (that is, a load device) is input. Therefore, the load switch **229** is connected to the power line **228**A. Further, the SW signal is at a high level, but the HP signal is at a low level. At time t**11**, the control circuit **2212** keeps clocking the set time.
(54) At time t**11**, the operator turns off the main power switch of the casing **21**. The DC voltage Vcc and the power supplied to the motor **226** start to decrease. The DC voltage Vdd also starts to decrease from 3.3 V toward 0 V. Since the connectors **216** and **224** are connected to each other, the HP signal remains at a low level. When the DC voltage Vdd starts to decrease, the control circuit

2212 detects that the main power switch is turned off or that connector **224** is removed from the connector **216**. In response to the detection, the control circuit **2212** starts outputting of the low-level SW signal, stops outputting of the PWM signal, and further initializes the timer ("0" in the drawing).
(55) FIG. **6** is a timing chart illustrating the operation when the pump assembly **22** is removed from the casing **21** in the refrigerant circulation device **2** illustrated in FIG. **1**.
(56) As illustrated in FIG. **6**, before time t**21**, the refrigerant circulation device **2** is in the same state as that before time t**11** (see FIG. **5**). At time t**21**, the operator starts to remove the pump assembly **22** from the accommodation space **215** (see FIG. **2**). As a result, the pump assembly **22** moves in the separation direction D**02** (see FIG. **2**) from the mounting position P**01**. In the process of moving in the separation direction D**02**, among the terminals **216**A to **216**C, first, the terminal **224**C is separated from the terminal **216**C, and the terminals **224**A and **224**B are separated from the terminals **216**A and **216**B, respectively, with a time difference. Therefore, the HP signal input to the terminal **2212**A transitions from the low level to the high level (that is, the DC voltage Vdd) in response to the terminal **224**C being separated from the terminal **216**C, and thereafter, maintains the high level until the terminal **224**A is separated from the terminal **216**A.
(57) After time t**21**, at time t**22**, when the terminals **224**A and **224**B start to be separated from the terminals **216**A and **216**B, respectively, the DC voltage Vcc and the power supplied to the motor **226** start to decrease. The DC voltage Vdd also starts to decrease from 3.3 V toward 0 V. Similarly, the HP signal begins to drop. When DC voltage Vdd starts to decrease, the control circuit **2212** detects that the main power switch is turned off or that the connector **224** is removed from the connector **216**. In response to the detection, the control circuit **2212** starts outputting of the low-level SW signal, stops outputting of the PWM signal, and further, initializes the timer ("0" in the drawing). The control circuit **2212** may start outputting of the low-level SW signal in response to the transition of the HP signal from the low level to the high level.
(58) The load switch **229** disconnects the power line **228**A in response to the transition of the SW signal from the high level to the low level. That is, the control circuit **2212** switches the load switch **229** from connection to disconnection in response to detection of removal of the connector **224** from the connector **216**. Therefore, the motor **226** can be stably stopped.
(59) FIG. **7** is a timing chart illustrating an operation of the pump assembly **22** when an operator attempts to remove the pump assembly **22** from the casing **21** and then reattaches the pump assembly **22** to the casing **21**.
(60) As illustrated in FIG. **7**, at time t**31**, the control circuit **2212** detects that the connector **224** is removed from the connector **216**, similarly to the case at time t**22** (see FIG. **6**). In response to the detection, the control circuit **2212** starts outputting of the low-level SW signal, stops outputting of the PWM signal, and further, initializes the timer. In addition, as the terminals **224**A and **224**B start to be separated from the terminals **216**A and **216**B, respectively, the DC voltage Vcc and the power supplied to the motor **226** start to decrease. The DC voltage Vdd also starts to decrease from 3.3 V toward 0 V. Similarly, the HP signal begins to drop.
(61) After time t**31**, at time t**32**, the operator moves the pump assembly **22** to the mounting position P**01** in the accommodation space **215** (see FIG. **2**). As a result, among the terminals **224**A, **224**B, and **224**C (see FIG. **3**), the terminals **224**A and **224**B are electrically connected to the terminals **216**A and **216**B, respectively. As a result, the control circuits **227** and **2212** start operation by the DC voltage Vdd generated by the power supply unit **2213**. In particular, the control circuit **2212** starts monitoring of the HP signal. When the DC voltage Vdd is input, the HP signal becomes a high level.
(62) After time t**32**, at time t**33**, the control circuit **2212** starts clocking by the timer with detection of conduction of the terminals **216**C and **224**C as a trigger. The control circuit **2212** periodically monitors the time measured by the timer.
(63) After time t**33**, at time t**34**, when the control circuit **2212** recognizes that the timer has reached

a set time ("20" in the drawing), the control circuit **2212** outputs a high-level SW signal from the terminal **2212**B. The drive unit **2211** gives a drive signal to the gate of the load switch **229** in response to the input of the high-level SW signal. Therefore, immediately after the drive signal is applied to the load switch **229**, a current flows through the capacitor of the protection circuit **2210** and the capacitor is charged as described with reference to FIG. **4**.
(64) A time at which the first time has elapsed from time t**33** is defined as t**35**. When the capacitor is fully charged or nearly fully charged at time t**35**, as described with reference to FIG. **4**, the load switch **229** is switched to connection, and the DC voltage Vcc is applied between the gate and the source of the MOSFET of the load switch **229** and to the terminals **225**A and **225**B of the drive unit **225**.
(65) Further, a time at which the second time has elapsed from time t**33** is defined as t**36**. At time t**36**, the control circuit **2212** starts to provide the PWM signal to the control circuit **227** as described with reference to FIG. **4**.
(66) The example embodiments of the present disclosure are described above with reference to the drawings. However, the present disclosure is not limited to the above example embodiment, and can be implemented in various modes without departing from the gist of the present disclosure. Additionally, the plurality of elements disclosed in the above example embodiment can be appropriately modified. For example, a certain element of all elements shown in a certain example embodiment may be added to a element of another example embodiment, or some elements of all elements shown in a certain example embodiment may be removed from the example embodiment.
(67) The drawings schematically show each element mainly in order to facilitate understanding of the present disclosure, and the thickness, length, number, interval, and the like of each element that is shown may be different from the actual ones for convenience of the drawings. The configuration of each element shown in the above example embodiment is an example and is not particularly limited, and it goes without saying that various modifications can be made without substantially departing from the effects of the present disclosure.
(68) The technology according to example embodiments of the present disclosure is suitable, for example, for cooling an electronic device.
(69) Features of the above-described example embodiments and the modifications thereof may be combined appropriately as long as no conflict arises.
(70) While example embodiments of the present disclosure have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present disclosure. The scope of the present disclosure, therefore, is to be determined solely by the following claims.

Claims

1. A control device comprising: a first connector attachable to and detachable from an external device; a power supply path to electrically connect a load device and the first connector; a load switch to switch between connection and disconnection of the power supply path; a protection circuit to protect the load switch from an inrush current to the load switch; and a control circuit; wherein the control circuit is operable to output a drive signal to switch the load switch from disconnection to connection after a preset set time as a time until a connection state between the external device and the first connector is stabilized, which has elapsed since connection of the first connector to the external device was detected, and the protection circuit includes at least a capacitor and switches the load switch from disconnection to connection for a predetermined first time as a time for gradually transitioning the load switch from OFF to ON based on a charging time of the capacitor by the drive signal.
2. The control device according to claim 1, wherein the protection circuit is operable to switch the load switch from disconnection to connection by inputting a charging voltage of the capacitor to

the load switch in the first time.
3. The control device according to claim 1, wherein the control circuit is operable to start outputting a drive signal to the load device when a second time longer than the first time elapses after the set time elapsed.
4. The control device according to claim 3, wherein the second time is shorter than the set time.
5. The control device according to claim 1, wherein the external device includes a second connector attachable to and detachable from the first connector; the second connector includes a second detection terminal to be grounded; the first connector includes a first detection terminal to be connected to the second detection terminal when the first connector is connected to the second connector; and the control circuit is operable to detect that the first connector is connected to the second connector on a basis of a voltage of the first detection terminal.
6. The control device according to claim 5, wherein the control circuit is operable to switch the load switch from connection to disconnection in response to detection of removal of the first connector from the second connector.
7. A pump assembly comprising: a first connector attachable to and detachable from an external device; a motor; a pump rotor that is rotatable by power generated by the motor; a power supply path electrically connected between the motor and the first connector; a load switch to switch between connection and disconnection of the power supply path; a protection circuit to protect the load switch from an inrush current to the load switch; and a control circuit to output a drive signal to switch the load switch from disconnection to connection after a preset set time as a time until a connection state between the external device and the first connector is stabilized, which has elapsed since connection of the first connector to the external device was detected; wherein the protection circuit includes at least a capacitor and switches the load switch from disconnection to connection for a predetermined first time as a time for gradually transitioning the load switch from OFF to ON based on a charging time of the capacitor by the drive signal, and the motor is operable to generate the power in response to the load switch being switched to connection.
8. A refrigerant circulation device comprising: the pump assembly according to claim 7; and a flow path through which refrigerant circulates by rotation of the pump rotor.